Lab 4 - Potpourri

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Introduction

In this lab you'll review and get practice with a variety of concepts, methods, and tools you've encountered thus far.

Part 1 - All about Quarto

Question 1

a. Add each of strings below to the code chunk provided in your document, render the document, and determine if the string is a proper code chunk label. If not, explain why and describe how you could fix it so the document renders.

   - Chunk label 1:

```
#| label: a-label
#| with-a-line-break
```

   - Chunk label 2:

```
#| label: areaaaaaaaaaaaaaaaaaaaaaalllllllllllllllllllyyyyyloooooooooooooooooooonglabel
```

   - Chunk label 3:

```
#| label: label with spaces
```

   - Chunk label 4:

```
#| label: label-with-dashes
```

> **Tip**
>
> Try each label option in the code chunk provided in your document. If it gives you an error (document doesn't render), you know it's not a proper chunk label.

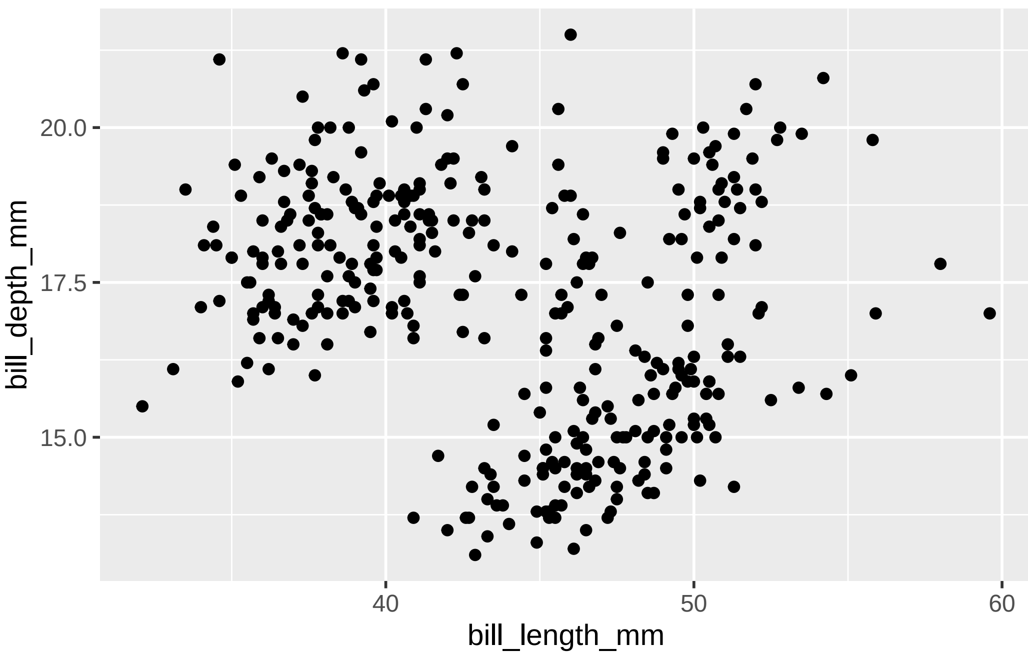
b. Which of the chunk label options above is the best option? Explain your reasoning in 1-2 sentences.

Question 2

You have the following code chunk:

```r
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



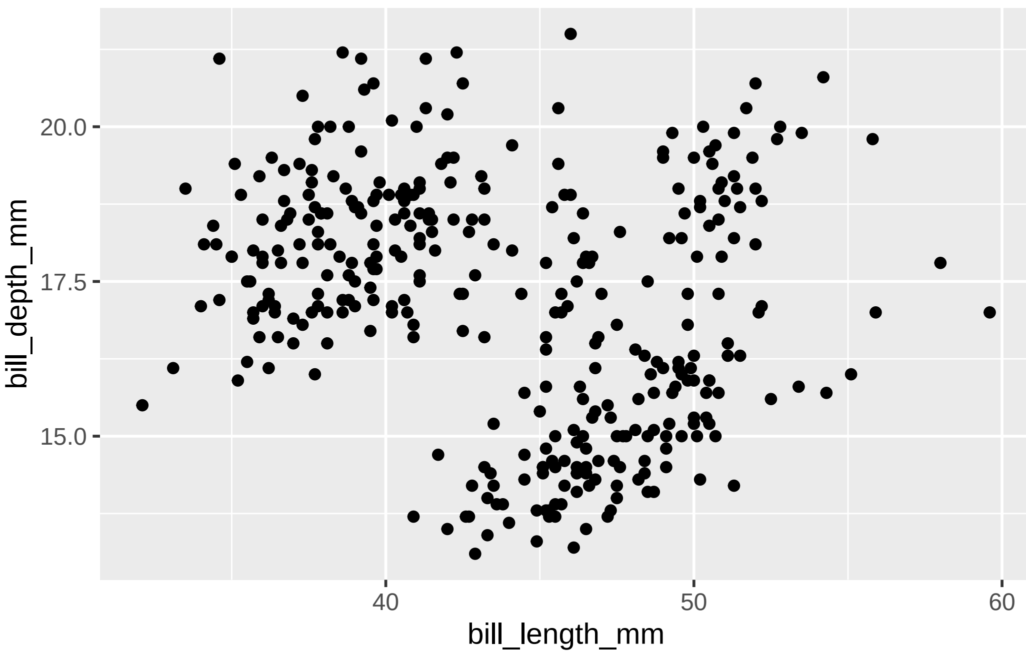
Add the following code chunk options, one at a time, and set each to `false` and then to `true`. After each value, render your document and observe its effect. Ultimately, choose the values that are the most appropriate for this code chunk. Based on the behaviors you observe, describe what each code chunk option does.

- echo
- warning
- eval

Echo True:

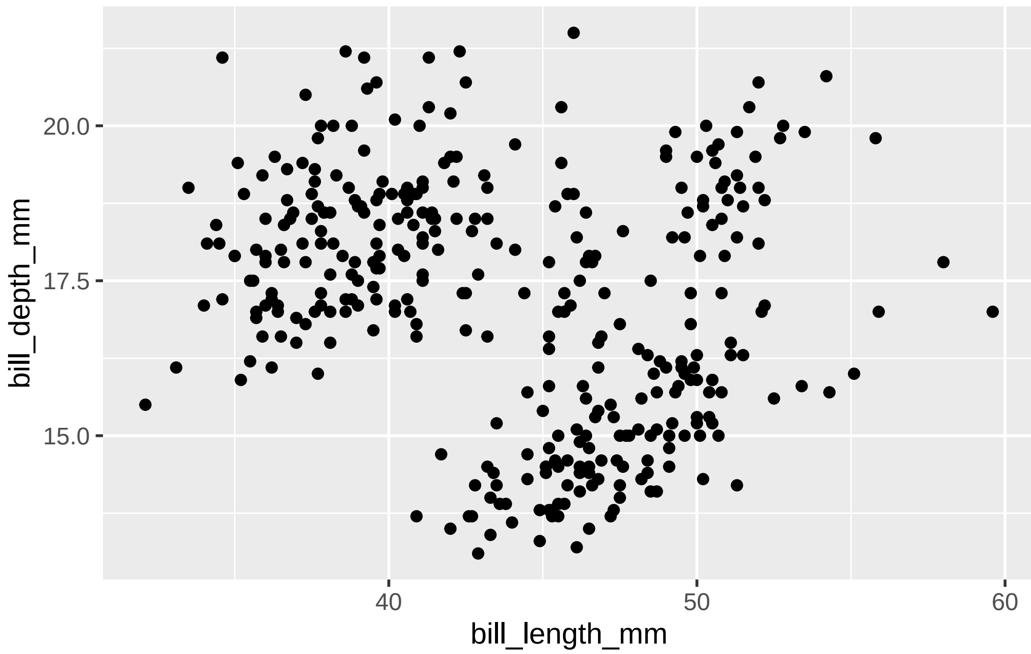
```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Echo False:

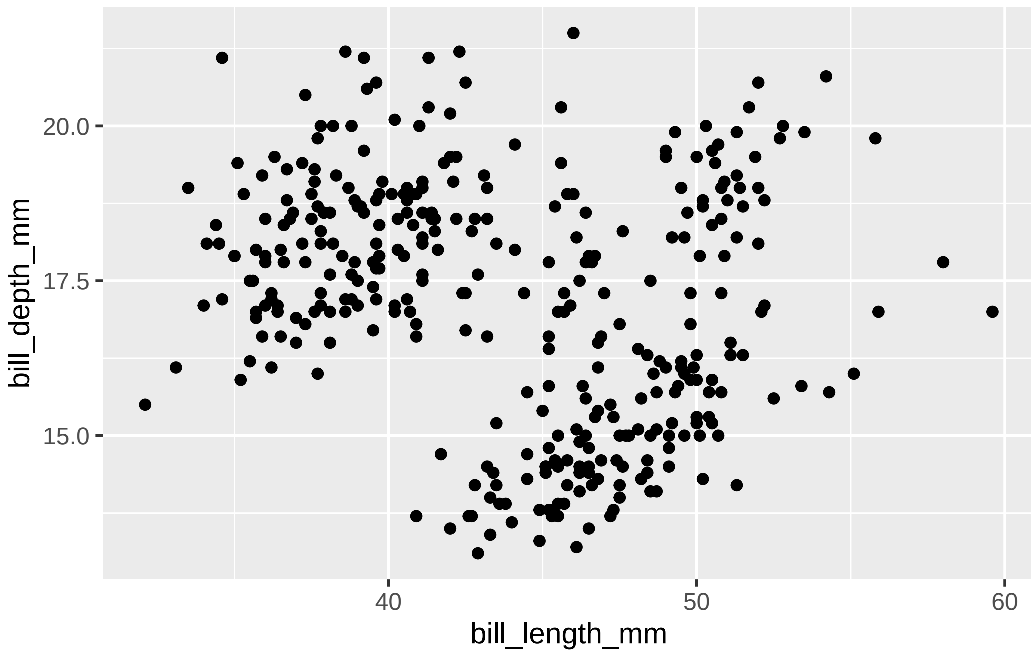
```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Warning True:

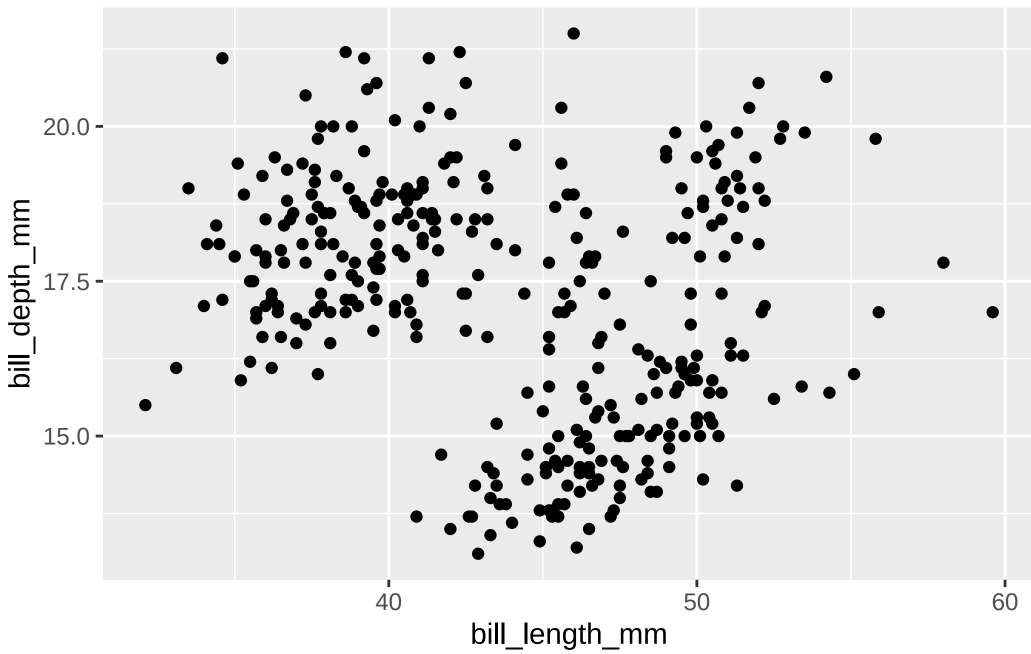
```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Warning False:

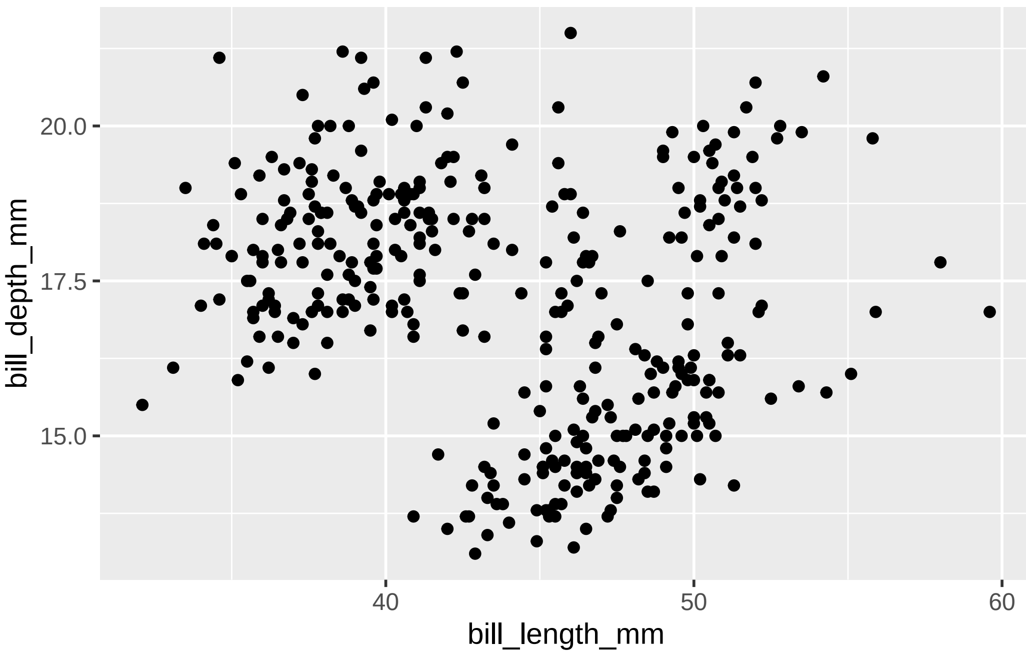
```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```



Eval True:

```r
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```

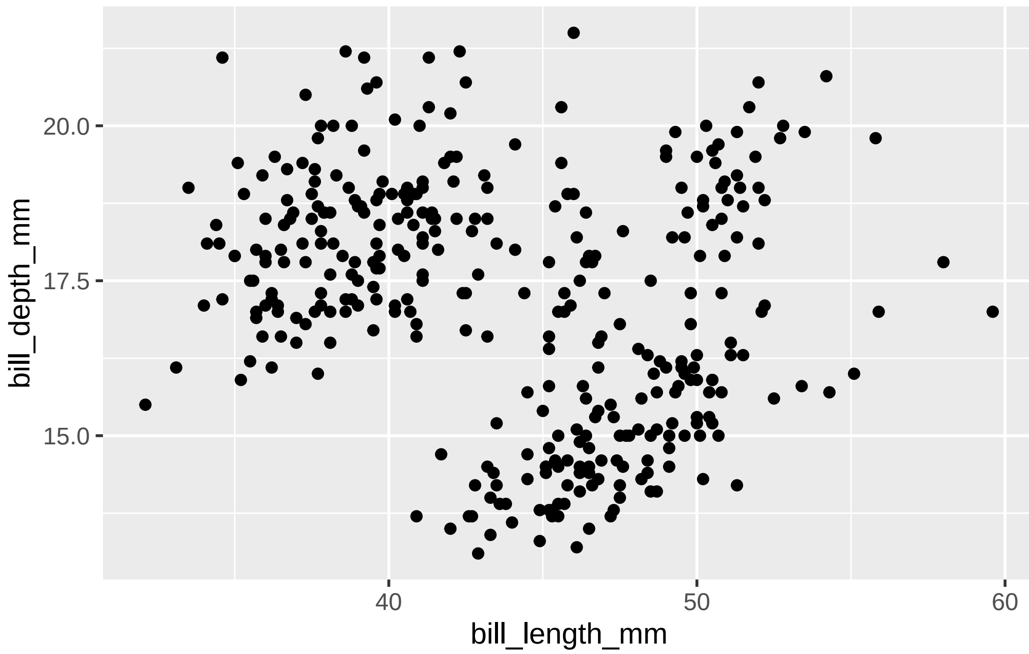


Eval False:

```r
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

Best Options: echo true, warning false, eval true

```r
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```



Comment:

What each one does:

Echo: decides if the code if shown in the rendered document (shown when true, absent when false)

Warning: decides if the warning messages are show in the output (shown when true, absent when false)

Eval: decides if the code is run or not in the rendered document (run when true, not run when false)

Question 3

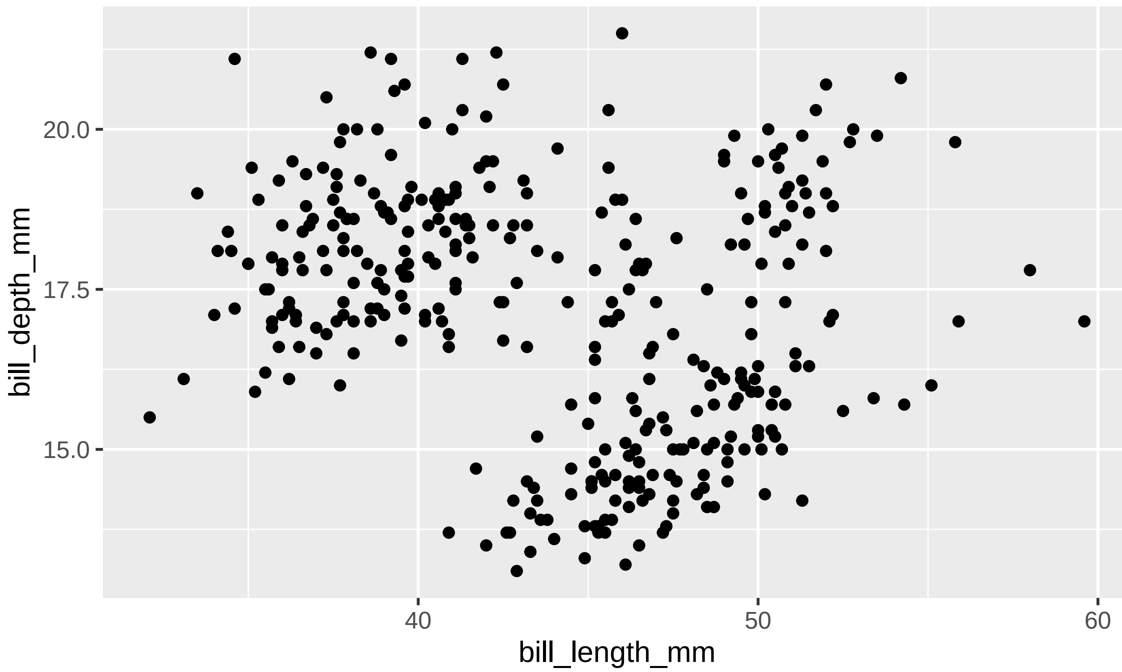
a. You have the following code chunk again.

```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

Add `fig-width` and `fig-asp` as code chunk options. Try setting `fig-width` to values between 1 and 10. Try setting `fig-asp` to values between 0.1 and 1. Re-render the document after each value and observe its effect. Ultimately, choose values that make the plot look visually pleasing in the rendered document. Based on the behavior you observe, describe what each chunk option does.

```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Comment:

Best options: width = 6 and asp = 0.6

What they do:

fig-width: controls the width of the plot (smaller value = thinner plot, larger value = wider plot)

fig-asp: controls the height of the plot relative to the width (smaller value = short wide plot, larger value = taller square plot)

a. b. You have the following code chunk, but look carefully, it's not exactly the same!

```
gplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

Add `error` as a code chunk option and set it to `false` and then set it to `true`. After each value, render your document and observe its effect. Ultimately, choose the value that allows you to render your document without altering the code. Based on the behavior you observe, describe what this code chunk option does.

```
gplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
  geom_point()
```

```
Error in `gplot()`:
! could not find function "gplot"
```

Comment: The best option is error: true because it allows for the document to still be rendered even though there is an error in the code.

> **Tip**
>
> Reading the documentation might also be helpful.

Render, commit, and push one last time. Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

Part 2 - Misrepresentation

Question 4

The following chart was shared by @GraphCrimes on X/Twitter on September 3, 2022.What is misleading about this graph?

(I had to remove the file link to the graph because it wasn't working)

a. Suppose you wanted to recreate this plot, with improvements to avoid its misleading pitfalls from part (a). You would obviously need the data from the survey in order to be able to do that. How many observations would this data have? How many variables (at least) should it have, and what should those variables be?
b. Load the data for this survey from `data/survation.csv`. Confirm that the data match the percentages from the visualization. That is, calculate the percentages of public sector, private sector, don't know for each of the services and check that they match the percentages from the plot.

Question 5

Create an improved version of the visualization. Your improved visualization:

- should also be a stacked bar chart with services on the y-axis, presented in the same order as the original plot, and services to create the segments of the plot, and presented in the same order as the original plot
- should have the same legend location
- should have the same title and caption
- does not need to have a bolded title or a gray background

How does the improved visualization look different than the original? Does it send a different message at a first glance?

> Tip
>
> Use `\n` to add a line break to your title. And note that since the title is very long, it might run off the page in your code. That's ok!
> Additionally, the colors used in the plot are `gray`, `#FF3205`, and `#006697`.

Render, commit, and push one last time. Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

Part 3 - DatasauRus

The data frame you will be working with in this part is called `datasaurus_dozen` and it's in the **datasauRus** package. This single data frame contains 13 datasets, designed to show us why data visualization is important and how summary statistics alone can be misleading. The different datasets are marked by the `dataset` variable, as shown in **?@fig-datasaurus**.

(I had to remove the image link again because it was causing errors when rendering)

> Note
>
> If it's confusing that the data frame is called `datasaurus_dozen` when it contains 13 datasets, you're not alone! Have you heard of a baker's dozen?

Here is a peek at the top 10 rows of the dataset:

```
datasaurus_dozen
```

```
# A tibble: 1,846 x 3
   dataset     x     y
   <chr>   <dbl> <dbl>
 1 dino     55.4  97.2
 2 dino     51.5  96.0
 3 dino     46.2  94.5
 4 dino     42.8  91.4
 5 dino     40.8  88.3
 6 dino     38.7  84.9
 7 dino     35.6  79.9
 8 dino     33.1  77.6
 9 dino     29.0  74.5
10 dino     26.2  71.4
# i 1,836 more rows
```

Question 6

In a single pipeline, calculate the mean of `x`, mean of `y`, standard deviation of `x`, standard deviation of `y`, and the correlation between `x` and `y` for each level of the `dataset` variable. Then, in 1-2 sentences, comment on how these summary statistics compare across groups (datasets).

> **Tip**
>
> There are 13 groups but `tibbles` only print out 10 rows by default. Add `print(n = 13)` as the last step of your pipeline to display all rows.

Question 7

Create a scatterplot of `y` versus `x` and color and facet it by `dataset`. Then, in 1-2 sentences, how these plots compare across groups (datasets). How does your response in this question compare to your response to the previous question and what does this say about using visualizations and summary statistics when getting to know a dataset?

> **Tip**
>
> When you both color and facet by the same variable, you'll end up with a redundant legend. Turn off the legend by adding `show.legend = FALSE` to the geom creating the legend.

Render, commit, and push one last time. Make sure that you commit and push all changed documents and your Git pane is completely empty before proceeding.

Grading

The lab is graded out of a total of 28 points.

You can earn up to 4 points on each question:

- 4: Response shows excellent understanding and addresses all or almost all of the rubric items.
- 3: Response shows good understanding and addresses most of the rubric items.
- 2: Response shows understanding and addresses a majority of the rubric items.
- 1: Response shows effort and misses many of the rubric items.
- 0: Response does not show sufficient effort or understanding and/or is largely incomplete.